

True bearings



1 State the true bearing of the following:

a Southwest

c West-Northwest

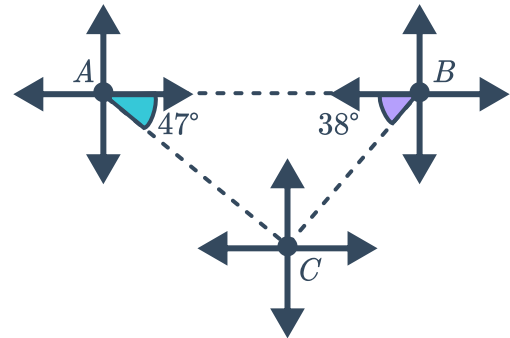
e Northeast

b Northwest

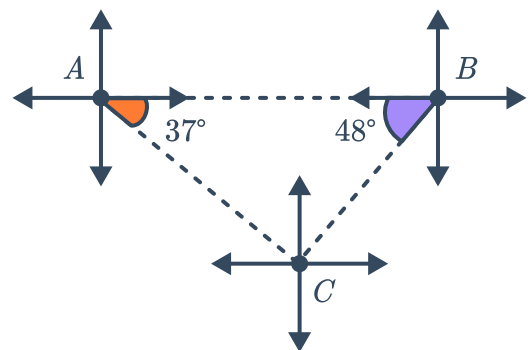
d South-Southeast

f Southeast

2 Determine the true bearing of point B from point A in the following bearings diagram:



3 Determine the true bearing of point A from point B in the following bearings diagram:



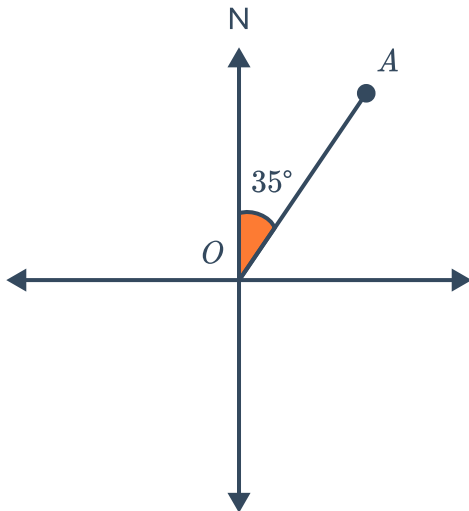


Compass bearings

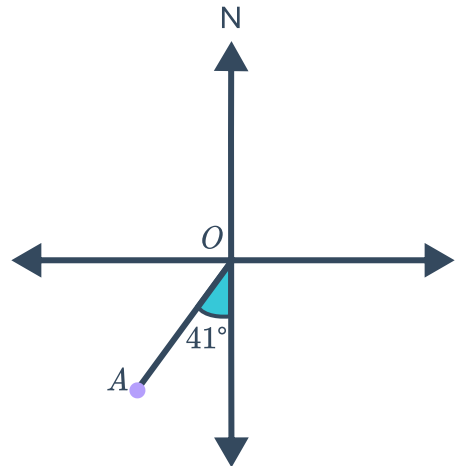
4 Consider the point A in the following diagrams:

- Determine the true bearing of point A from O .
- Find the compass bearing of point A from O .

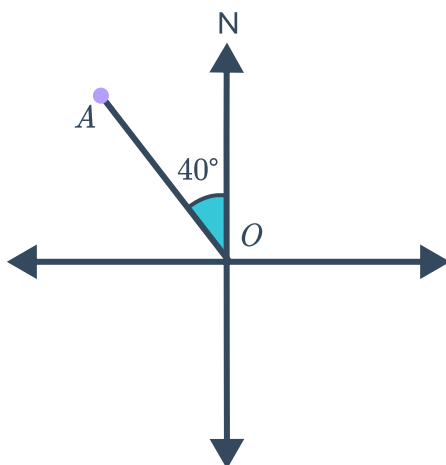
a



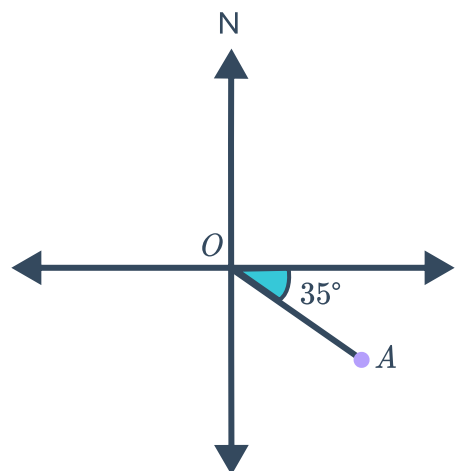
b



c

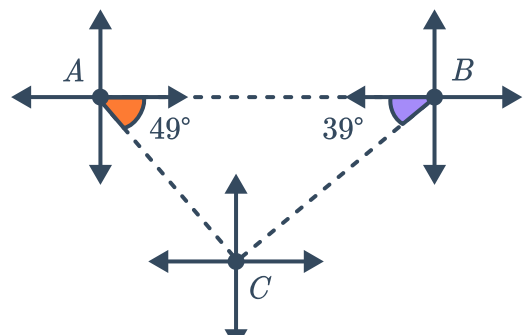


d



5 In the figure below, point B is due East of point A :

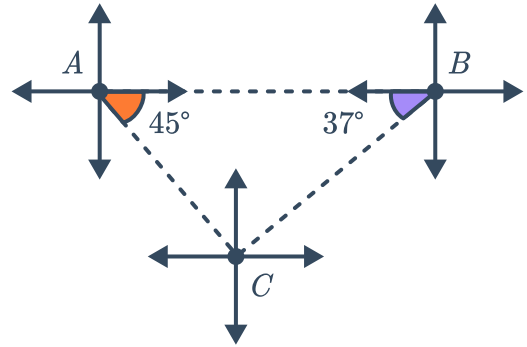
- Find the true bearing of point A from point C .
- Find the compass bearing of point A from point C .





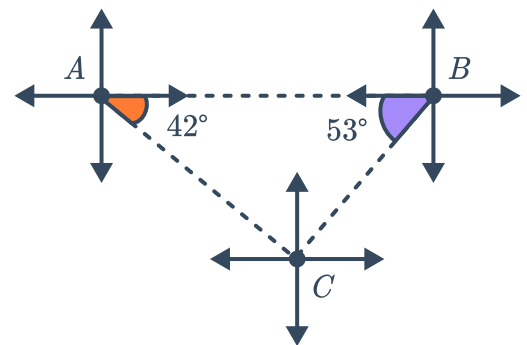
6 Consider the following diagram:

- a Find the true bearing of point B from point C .
- b Find the compass bearing of point B from point C .



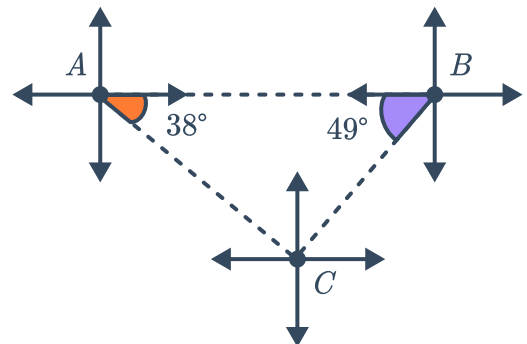
7 Consider the following diagram:

- a Find the true bearing of point C from point A .
- b Find the compass bearing of point C from point A .



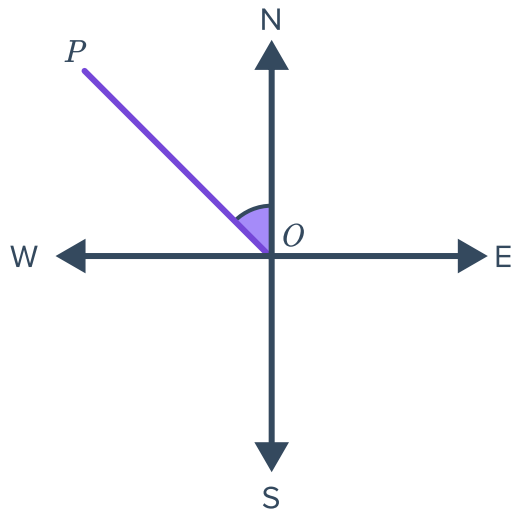
8 Consider the following diagram:

- a Find the true bearing of point C from point B .
- b Find the compass bearing of point C from point B .

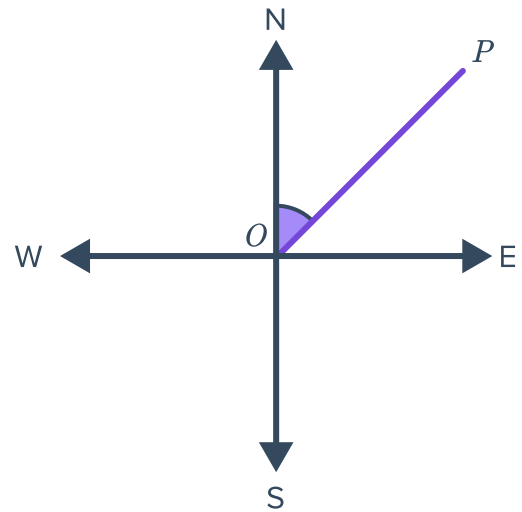


- 9 Determine whether the following diagrams  lay the point P with a compass bearing between N and W from the point O .

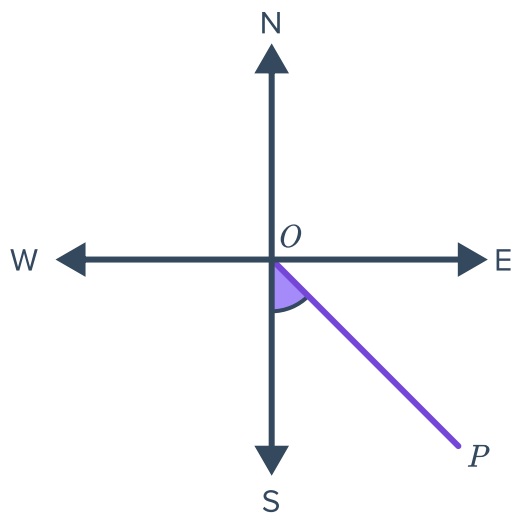
a



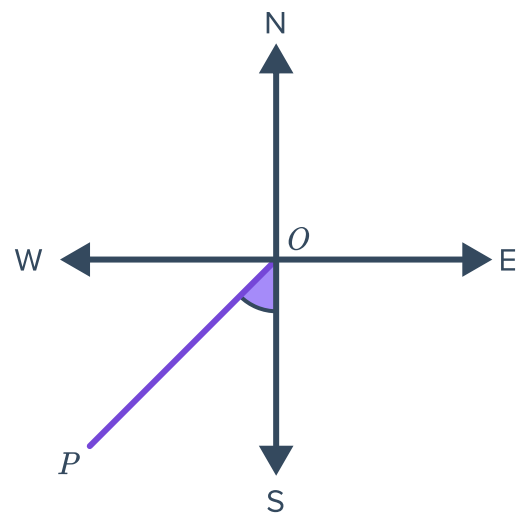
b



c

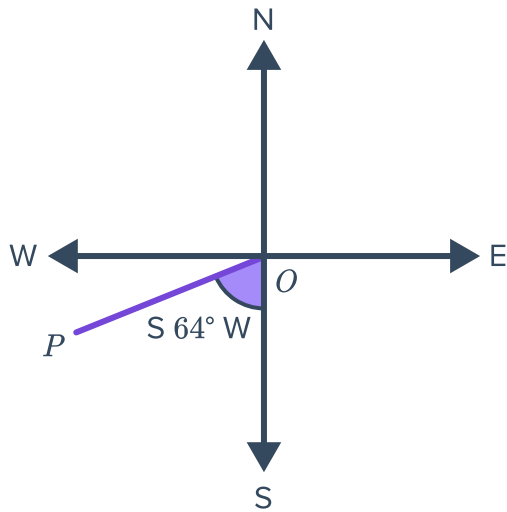


d

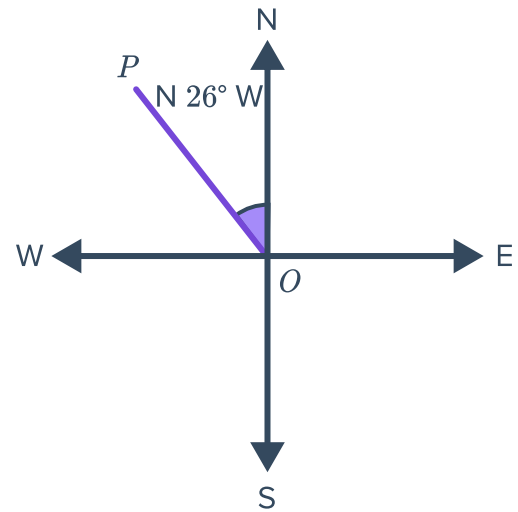


- 10 Determine whether the following diagrams show a true bearing of 296° T for the point P from the origin O .

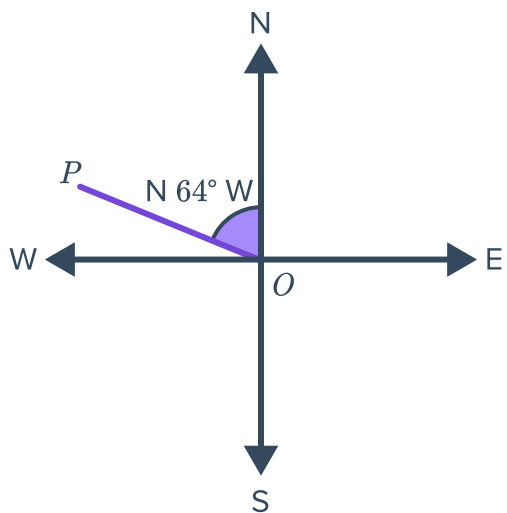
a



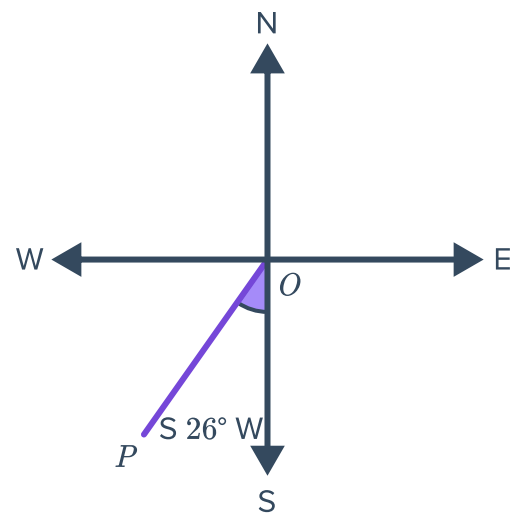
b



c

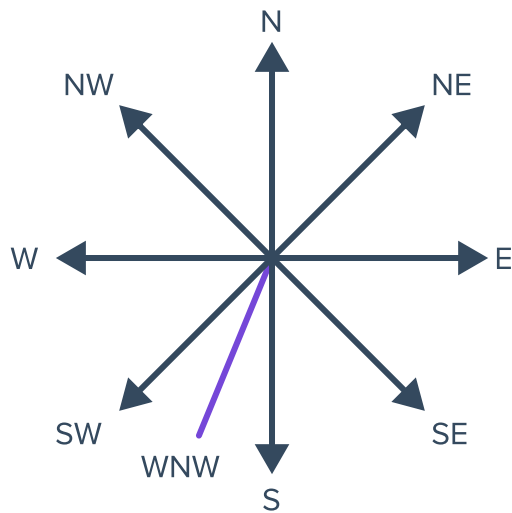


d

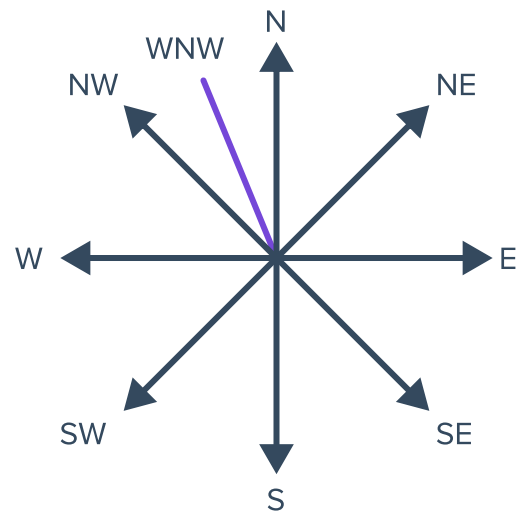


11 Determine whether the following diagram  lay *WNW*:

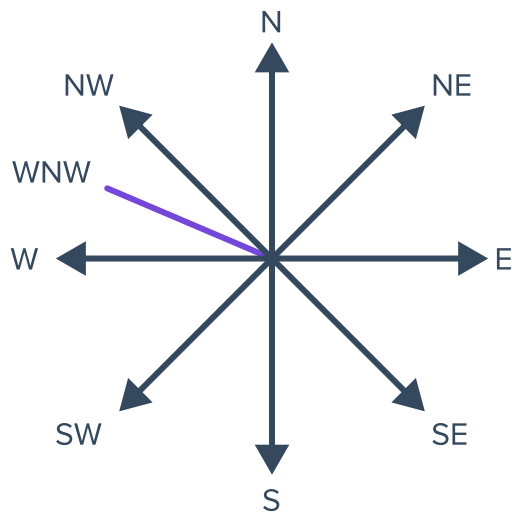
a



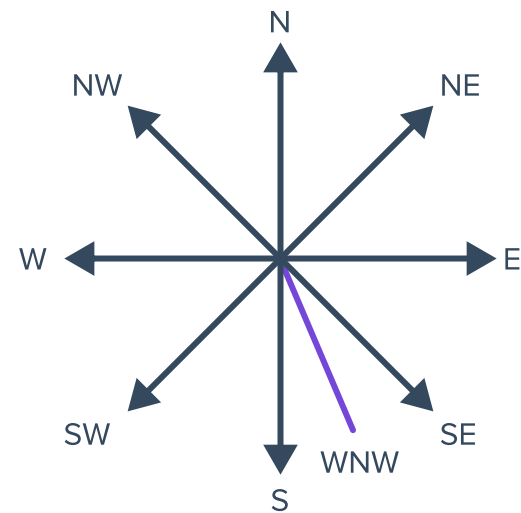
b



c



d

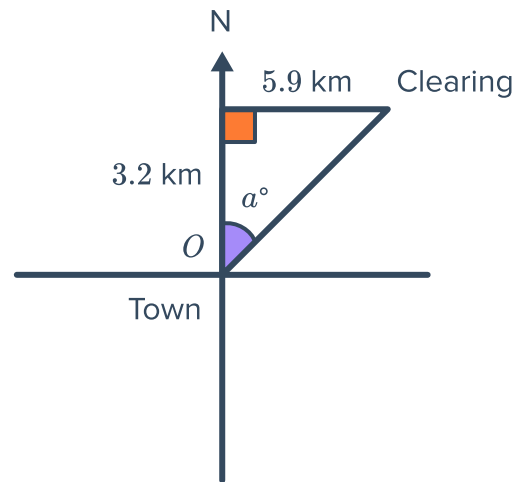


12 If a person walks at a compass bearing of *SW*, and then turns around to go back in the direction they came, what direction are they heading now?



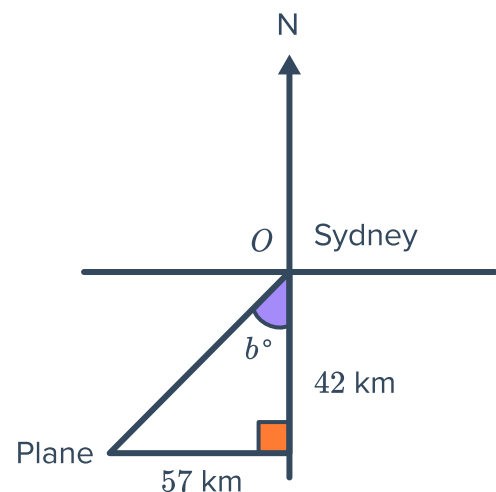
Applications

- 13** Consider the following diagram:
If the bearing of the clearing from the town is a° , find a to the nearest degree.



- 14** An aeroplane sets off on a bearing of 051°T . After some time, it needs to turn back and head to its original position. On what true bearing must it travel?
- 15** A boat sets off on a bearing of 125°T . After some time, it needs to turn back and head to its original position. On what true bearing must it travel?
- 16** Shortly after take-off, a plane is 42 km south and 57 km west of the airport in Sydney that it left from:

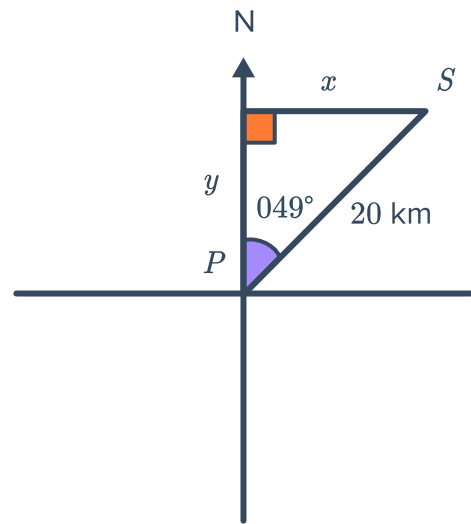
- a** Find the size of the angle marked b , to one decimal place.
- b** Hence, find the compass bearing of the plane from the airport.
- c** State the true bearing of the plane from the airport.



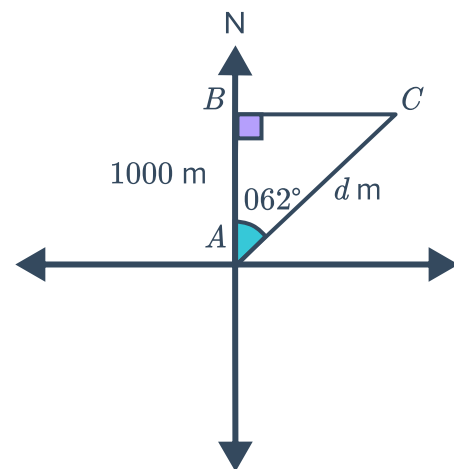
- 17 The position of a ship S is given to be 20 km from P , on a true bearing of 049° T. The position of the ship can also be given by its (x, y) coordinates.



- Find the ship's x -coordinate to one decimal place.
- Find the ship's y -coordinate to one decimal place.

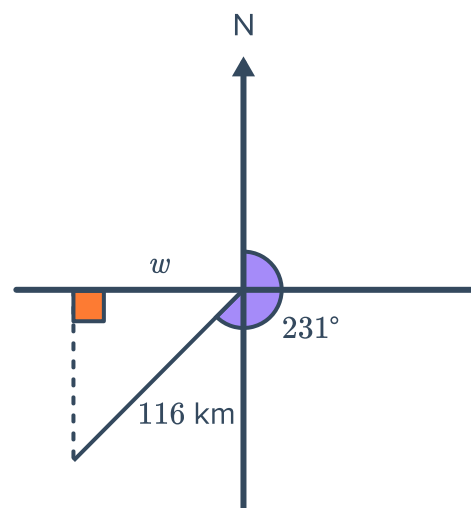


- 18 On an orienteering course, Beth runs 1000 metres north from point A to point B , then turns east and runs to point C . If the true bearing of C from A is 062° T and the direct distance Beth must run to get back to point A is d metres, find d to the nearest metre.



- 19 During a group hike, a hiker walked 11 km NE of base camp. In the evening the hiker was separated from the group and after a night lost in the wilderness, the hiker was found 11 km NW of the base camp. What compass bearing must the hiker have followed during the night?

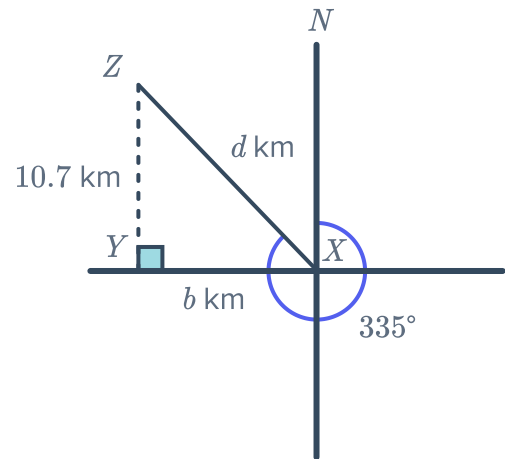
- 20 Luke sailed for 116 km on a bearing of 231° . If w is how many kilometres west he has sailed from his starting point, find w to one decimal place.



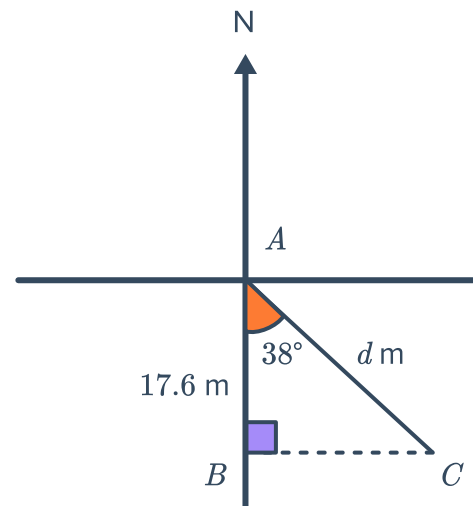


- 21** During a rescue search, a helicopter flew west from point X to point Y , then changed course and flew 10.7 km north to point Z . If point Z is on a bearing of 335° T from point X :

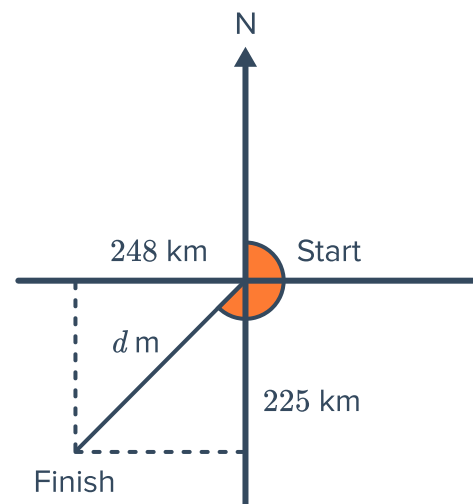
- Find the size of $\angle YXZ$.
- If the distance from point Y to point X is b km, find b to one decimal place.
- If the distance that the helicopter must fly between point Z and point X is d km, calculate d to one decimal place.



- 22** Three television presenters are practising their navigation skills before heading off on an expedition to a remote location. Belinda at point B is positioned 17.6 m south of Amelia at point A . Carl at point C is due east of Belinda and on a bearing of $S\ 38^\circ\ E$ from Amelia. If Amelia and Carl are d m apart, find d to one decimal place.



- 23** A yacht sailed in a direction so that its final position was 248 km west and 225 km south of its starting point.
- If the true bearing on which the yacht sailed is b° , find the value of b to one decimal place.
 - If the boat has sailed a total of d m, find the value of d to one decimal place.



24 In remote locations, photographers must keep track of their position from their base. One morning a photographer sets out from base, represented by point B , to the edge of an ice shelf at point S on a bearing of 055° . She then walked on a bearing of 145° to point P , which is 916 m due east of base.

- a State the size of $\angle BSP$.
- b Find the distance BS , to one decimal place.
- c Find the distance SP , to one decimal place.
- d If the photographer walks directly back to her base from point P , determine the total distance she would have travelled. Round your answer to one decimal place.

